

SIGNAL VERIFICATION IN THE REAL WORLD: WHEN IS A SIGNAL NOT A SIGNAL?

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Abstract—The confirmed reception of an ETI signal would be very important to the entire Earth and is, consequently, an issue about which emotions run high. The elusive ETI signal is likely to be at the very edge of detectability and its very existence may be doubtful. The procedure for verifying the reality of our first ETI signal is of central importance. Public announcements which later turn out to be false alarms should be avoided.

To help inject a healthy skepticism into our work, I propose that the following three principles should be adhered to by all ETI searchers who use real time data:

1. Be skeptical! Verification must involve outside scientific peer review and independent confirmation by a second telescope.
2. Show restraint! Public announcements should be made only after the signal is verified to be of extraterrestrial origin.
3. Plan ahead! A procedure for rapid verification of ETI signals and dissemination of ETI data through normal scientific channels should be established before searching begins.

1. INTRODUCTION

Any discussion of the legal, political and social implications of detecting extraterrestrial intelligence has to be set in the context of the difficulty of ascertaining and certifying the reality of a suspected ETI signal. The importance of receiving a confirmed ETI signal can not be overstated. Such an event would affect the entire Earth and is, consequently, an issue that raises strong emotions. The experience of astronomers in searching for new phenomena indicates that an elusive ETI signal may well be at the very edge of detectability. The signals may either be faint, as would be the case for unintentional leakage radiation, or of short duration, as would be the case if the message were part of a test or calibration. In either case, the certainty of a suspected ETI signal may be doubtful and the verification procedure is of central importance.

A public announcement of detection which later is proven to be either a false alarm or a hoax should be avoided. Such public false alarms would be very detrimental to the continuation of serious SETI efforts not only by casting doubt upon SETI as science, but also by encouraging a rash of unfounded claims and deliberate hoaxes. To keep our current search programs alive, we must display the highest ideals of scientific responsibility and show the utmost restraint.

2. THE PROBLEM OF FINDING FAINT SIGNALS

As in all endeavors where we are passive observers of the signals which fall upon the Earth, the problem with detecting an ETI signal will be one of deciding when we are sure we

have detected a real signal. Of course we may be lucky and find ourselves in the middle of a strong, narrowly beamed signal which originates from a nearby star. In this case the signal will probably be unmistakable.

However, given the enormous distances between stars it is quite likely that any ETI signal will be very weak and nearly indistinguishable from the natural noise surrounding it. Astronomers have a great deal of experience in dealing with weak signals. As astronomers we are constantly reminded of how difficult and time-consuming it is to assess the reality of and to accurately measure spectral features which are near the limit of detectability.

The other problem is that of finding a signal of short duration which ceases before it can be properly analyzed and confirmed to be of extraterrestrial origin. A strong signal which tantalized us in this fashion would be most difficult to deal with. In any case we should not expect to have an easy time confirming a suspected ETI signal.

3. AN EXAMPLE FROM ASTRONOMY

One current example illustrates the difficulties of confirming a discovery which, if real, will be critically important to our understanding of the origin of our universe. The discussion is about whether the two quasar images Q1146-111 B, C are the result of a supermassive gravitational lens. One would think that confirmation of a discovery dealing with such a fundamentally important topic would be verified rapidly. The real world is, alas, not as cooperative as we might like. A short recap of the controversy is instructive.

Turner[1] from Princeton is the leader of a group which says the spectra of the two images are identical - indicating that both objects are images of the same quasar, formed by an exceptionally strong gravitational lens. What makes this a special case is the wide separation of the two images, thus requiring an extraordinarily massive object to act as the lens. But we see nothing there to act as a lens, implying that the object must be a form of matter or energy with which we have no experience. The most widely favored explanation invokes the presence of an infinitesimally thin, but extremely massive, cord of energy known as a cosmic string. This is the first possible evidence for a cosmic string that has been found. Confirmation of the presence of such a cosmic string would have far-reaching consequences for our understanding of the Big Bang and the origin of the universe.

On the other hand, Shaver and Cristiani[2] of the European Southern Observatory maintain that their spectra of the same two objects show substantial differences which, they say, rule out the gravitational lens explanation. The whole picture is complicated by the fact that quasar spectra are known to vary as time passes, so that, according to Turner, differences do not rule out the gravitational lens interpretation. In a final complication, Phinney and Blandford[3] of Caltech argue that the two quasars might be a random, chance association and really have nothing to do with each other.

The recent developments are nicely summarized by Waldrop[4,5] who concludes that the picture remains uncertain and that the only way to make further progress is by obtaining more observations. Unfortunately, the quasar in question is now behind the Sun and will not be available for observation for another six months.

This discussion is, as scientific controversies go, a relatively heated one, and has been carried out in the newspapers as well as in the scientific literature and on the floor at scientific meetings. Nevertheless, the spirit of the process has been one of scientific inquiry. Challenges to the original discovery have been based on new observations or new calculations and every effort is being made to uncover additional evidence which can help to resolve the issue. Information about the observations has been disseminated widely within the astronomical community and confirming observations have been obtained rapidly, at least on the normal time scale of astronomical research. Nevertheless, a definitive answer as to whether or not a cosmic string has been discovered will

not be available until at least a year after the initial observations were made, and it may take much longer. As of now we do not even know if the apparent differences in the spectra of the two objects are real or not.

What lessons can we learn from the present cosmic string controversy which can be applied to the reception of a possible ETI signal:

1. Weak signals are difficult to detect and measure unambiguously.
2. Independent confirmation at a second observatory of signals which vary with time is impossible to do with certainty unless done concurrently with the initial observations.
3. Rapid and open dissemination of observations among the scientific community will result in a constructive, critical discussion and will spur attempts by other scientists to make confirming observations.
4. Acceptance of the reality of a new type of phenomenon in the scientific community will come only after a critical and objective review of all the evidence at hand by knowledgeable scientific peers. Such acceptance takes considerable time.
5. During the process of confirmation there is no clearly apparent instant at which a signal becomes "confirmed." Confirmation occurs gradually as more and more scientists become convinced by the evidence.

4. THE VERIFICATION PROCESS

The scientific approach to verification is a far cry from assembling a jury and deciding upon guilt or innocence. The scientist does not make a final decision, but instead adopts a "working hypothesis" while still continuing to collect evidence, modifying his or her hypothesis on the basis of additional observations. This style causes a classic problem whenever scientists are asked to provide advice to policy makers. Such an open ended approach will also cause difficulties in the case of an ETI signal, where we are considering establishing major policies for announcing the first signal and are wrestling with legal problems which will confront us when we do detect a signal. Yet the most difficult problem is likely to be the determination of when a suspected signal should be accepted as real.

In addition to the problems which face the astronomer seeking to confirm a new discovery, there are three factors which add nearly insuperable complications to the confirmation of a suspected ETI signal:

1. Discovery of an ETI signal involves not only the scientific community but the whole world.
2. ETI is a very emotional subject and the pressure for an immediate confirmation will be enormous.
3. ETI has been, and will continue to be, the subject of both intentional and unintentional hoaxes.

We can ease the problems of communication between scientist and policy maker in the case of determining the reality of our first ETI signal if all parties understand that there may be difficulties in verifying the signal and if we all agree beforehand to a reasonably stringent criterion for detection and a responsible style of behavior in making the announcement.

5. RECOMMENDATIONS

It is clear that we in the SETI business must prepare ourselves for the eventual reception of a suspect signal which passes all the immediate tests done on the spot by the discovery team and which looks like a real signal of extraterrestrial origin. But at that point, how do we proceed with verification? At what level of verification will the SETI community agree that the signal is real? When should a public announcement be made? I remind us all of the necessity of establishing an on-site testing procedure and propose the following recommendations which speak to these points.

On-Site Testing Procedure: Every search program worthy of the name must have a series of well thought out checks designed to test the reality of any suspected signal. These should be written down and left with every telescope operator, or should be incorporated into search software. They should include tests such as scanning that part of the sky to locate the signal source, changing the local oscillator frequency, and scanning across the spectrum. Polarization checks and other tests which may be dependent upon the characteristics of the specific instrumentation at the site should be included.

Recommendation 1: Anyone involved in SETI activities should be urged to adopt as a fundamental criterion that no signal will be accepted as real until it is confirmed by independent observations at a second site and the data has been subjected to an outside scientific review. The importance of the discovery of ETI to the entire world warrants this high degree of skepticism. It would be helpful to publicize this criterion widely so as to alert the press and the interested public to the

difference between suspected signals and confirmed signals.

Recommendation 2: All ETI searchers bear a responsibility for avoiding premature announcements and false alarms. Owing to the fundamental importance and highly emotional nature of discovering a true ETI signal, public announcements should be avoided until the reality of the suspected signal is confirmed. While this may be difficult to accomplish completely, it is desirable to get as much of the confirmation process completed before the world's press becomes involved.

Recommendation 3: Every search program should include a plan for rapid dissemination of all relevant technical information about suspected signals to the scientific community. If possible, tentative plans should be made for the confirmation of such signals, which observatories to call, who to invite in to scrutinize the data, etc.

The advantage of the process outlined above is that it mimics the standard approach which has served the field of astronomy since the time of Galileo, with various degrees of success. The crucial aspect is rapid dissemination of information by the discoverer and the ability of other astronomers at different observatories to repeat the observations. The problem faced by SETI scientists is that the normal time scale for the confirmation of observations will be many times shorter than we, as astronomers, are used to. Determining the reality of a suspected ETI signal will not be easy in any case, but planning ahead will ease the problems.

REFERENCES

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EDITORS' NOTE

Since this paper was presented, new observations of Q1146-111B,C have been made.

The situation is not yet completely clear but it now seems more likely that these are two separate quasars, not two images of the same quasar. The fact that questions still remain two years after the initial publication emphasizes the author's point about the difficulty of verifying a weak signal.